# Reflexive, Symmetric, and Transitive Relations

A relation $R$ on a set $A$ is ***reflexive*** if, and only if, *every element* of $A$ is related to itself by $R$. That is, $\forall x \in A\ (x\,R\,x)$. `for all x ∈ A, (x R x)`

$R$ is ***irreflexive*** if, and only if, *no* element is related to itself: $\forall x \in A\ \neg(x\,R\,x)$.

`for all x ∈ A, ¬(x R x)`

A relation $R$ on a set $A$ is ***symmetric*** if, and only if, $y$ is related to $x$ whenever $x$ is related to $y$. That is, $\forall\, x, y \in A \Big((x\,R\,y) \Rightarrow (y\,R\,x)\Big)$.

`for all x, y ∈ A, (x R y) ⇒ (y R x)`

$R$ is ***antisymmetric*** if, and only if, $y$ is *never* related to $x$ if $x$ is related to $y$, *except* possibly for when $x{=}y$. That is, $\forall\, x, y \in A \Big((x\,R\,y) \wedge (y\,R\,x) \Rightarrow x{=}y\Big)$.

`for all x, y ∈ A, [(x R y) ∧ (y R x) ⇒ x = y]`

$R$ is ***transitive*** if, and only if, $x$ is related to $z$ whenever $x$ is related to some $y$ which is related to $z$. That is, $\forall\, x, y, z \in A \Big((x\,R\,y) \wedge (y\,R\,z) \Rightarrow (x\,R\,z)\Big)$.

`for all x, y, z ∈ A, [(x R y) ∧ (y R z) ⇒ x R z]`

# Reflexive, Symmetric, and Transitive Relations

**Examples:**

| | $=$ | $<$ | $\leq$ | *sibling* |
|---|---|---|---|---|
| reflexive | YES | NO | YES | NO |
| irreflexive | NO | YES | NO | YES |
| symmetric | YES | NO | NO | YES |
| antisymmetric | YES | YES | YES | NO |
| transitive | YES | YES | YES | NO |

# Reflexive, Symmetric and Transitive Relations

**Exercise:** Let $A = \{$ cat, dog, bird, rat $\}$, and let

$$R_A = \{ (x, y) : \ x \text{ and } y \text{ have a letter in common } \}$$

Draw the relation $R_A$, and determine if it is:

- reflexive
- irreflexive
- symmetric
- antisymmetric
- transitive

**Exercise:** Is the *is-an-ancestor-of* relation over people:

- reflexive ?
- irreflexive ?
- symmetric ?
- antisymmetric ?
- transitive ?

# Orderings Relations

A relation $\mathrm{R}$ on a set $\mathrm{A}$ is a ***partial order*** if, and only if, $\mathrm{R}$ is reflexive, antisymmetric and transitive.

$\mathrm{R}$ is a ***total order*** if, and only if, $\mathrm{R}$ is a partial order in which any two elements are related in one way or the other: $\forall\, x, y \in A \left( (x \,\mathrm{R}\, y) \vee (y \,\mathrm{R}\, x) \right)$.

**Example:**

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for all x, y ∈ A, [(x R y) ∨ (y R x)].
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- $=$ on $\mathbb{Z}$ is a partial order but not a total order.
- $\leq$ on $\mathbb{Z}$ is a total order.
- $<$ on $\mathbb{Z}$ is not a partial order as it is not reflexive.
- $\subseteq$ on sets is a partial order but not a total order.

# Orderings Relations

**Exercise:** Which of the following binary relations on $\mathbb{N}$ are partial orders? Which are total orders?

1. The identity relation $\mathrm{I} = \{ (n, n) : n \in \mathbb{N} \}$.

2. The universal relation $\mathrm{U} = \{ (m, n) : m, n \in \mathbb{N} \}$.

3. The parity relation $\mathrm{P} = \{ (m, n) : m \equiv n \pmod 2 \}$.

# Equivalence Relations

A binary relation $\mathrm{R}$ on a set is an ***equivalence relation*** if, and only if, $\mathrm{R}$ is reflexive, symmetric and transitive.

**Example:**

- $=$ on $\mathbb{Z}$ is an equivalence relation.
- $\leq$ on $\mathbb{Z}$ is not an equivalence relation as it is not symmetric.
- $<$ on $\mathbb{Z}$ is not an equivalence relation as it is not symmetric and not reflexive.
- $\subseteq$ on sets is not an equivalence relation, as it is not symmetric.
- the *same-month-of-birth* relation on people – relating two people if, and only if, their birthdays are in the same month – is an equivalence relation.

# Equivalence Relations

**Exercise:** Which of the following binary relations on $\mathbb{N}$ are equivalences?

1. The identity relation $I = \{ (n, n) : n \in \mathbb{N} \}$.

2. The universal relation $U = \{ (m, n) : m, n \in \mathbb{N} \}$.

3. The parity relation $P = \{ (m, n) : m \equiv n \pmod 2 \}$.

# Equivalence Classes and Partitions

Consider the equivalence relation *same-month-of-birth*:

$$\{\, (x, y) : \text{ } x \text{ and } y \text{ have birthdays in the same month } \}.$$

This equivalence relation partitions a set $A$ of people into 12 groups depending on their birthday. For example, one group might consist of all those who have their birthday in November. There may be fewer than 12 groups, if there are months in which no one in $A$ was born.

A ***partition*** of a set $A$ is a collection $\{\, A_i : i \in I \,\}$ of disjoint non-empty subsets of $A$, which together contain all of $A$. That is:

$$A_i \cap A_j = \emptyset \quad \text{whenever } i \neq j; \quad \text{and} \quad \forall x\, (x \in A \Leftrightarrow \exists i \in I\, x \in A_i).$$

```
for all x (x ∈ A ⇔ ∃ i ∈ I, x ∈ Aᵢ)
```

Each $A_i$ is called a ***block*** of the partition.

**Example:** Over the set $A = \{1, 2, 3, 4\}$,

- $\{\{1, 2\}, \{3, 4\}\}$ and $\{\{1, 3, 4\}, \{2\}\}$ are partitions of $A$.
- $\{\{1\}, \{3, 4\}\}$ and $\{\{1, 2\}, \{2, 3, 4\}\}$ are *not* partitions of $A$.

# Equivalence Classes and Partitions

**Exercise:** Which of the following form a partition of the set $A = \{a, b, c, d\}$?

1. $\{\{a\}, \{b, c\}, \{d\}\}$
2. $\{\{a, c\}, \{b, d\}\}$
3. $\{\{a, c\}, \{b\}\}$
4. $\{\{a, b, c, d\}\}$
5. $\{\{a, c\}, \{b\}, \{c, d\}\}$
6. $\{\{a\}, \{b\}, \{c\}, \{d\}\}$

**Exercise:** How many partitions are there on the set $\{1, 2, 3\}$?

# Partitions and Equivalence Relations

A partition defines an equivalence relation in the following way: two elements are related if, and only if, they appear in the same block of the partition.

Conversely, as we've seen, any equivalence relation on a set partitions the set into disjoint non-empty subsets (blocks), called the ***equivalence classes***.

Given an equivalence relation $R$ on $A$, the *equivalence class* of $x \in A$ with respect to $R$, denoted $[x]_R$, is the set of elements of $A$ related to $x$ by $R$:

$$[x]_R = \{\, y \in A : x \,R\, y \,\}.$$

The collection $\{\, [x]_R : x \in A \,\}$ of these equivalence classes is a partition of $A$.

We say that one partition is a ***refinement*** of a second partition if, and only if, every block of the first is a subset of some block of the second.

**Example:** We can refine the *same-month-of-birth* relation into a *same-month-of-birth-and-same-gender* relation by also splitting the people in $A$ according to gender. Thus for example, one group might consist of all females whose birthday is in November.